# ECON 204 – Contemporary Economic Issues: In-Class Empirical Activity

**Format: groups of 2–3** **Time: 45–60 minutes**

**Tool: Excel or Google Sheets**

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**Learning Objectives:**
In this activity you will (i) construct Canadian inflation from CPI data, (ii) connect inflation to cost-of-living changes, and (iii) evaluate how the Bank of Canada's policy rate moves relative to inflation.

## Part A. Organize the dataset (10 minutes)

Open the Excel template provided by the instructor. Confirm that the `Data` sheet contains these columns: `date`, `cpi`, `policy_rate`, `avg_hourly_earnings`.

**Checkpoint A1.** What is the first month in the dataset? What is the last month?

## Part B. Construct inflation (10–15 minutes)

Use the CPI series to compute year-over-year inflation:

$$\pi_t = \left(\frac{CPI_t - CPI_{t-12}}{CPI_{t-12}}\right) \times 100.$$

In the template, this is computed on the `Calculations` sheet.

**Checkpoint B1.** Identify the month with the highest year-over-year inflation in 2015–2025. Record the month and the inflation rate.

## Part C. Make two charts (10–15 minutes)

Create the following two charts (titles and axis labels matter):

- **Chart 1:** CPI level and inflation (YoY). If you use a secondary axis, explain what each axis measures.
- **Chart 2:** Inflation (YoY) and the policy rate. Use the same monthly date axis.

**Checkpoint C1.** In Chart 2, does the policy rate appear to rise *before* inflation peaks, *after* inflation peaks, or roughly *at the same time*? Give one sentence.

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## Part D. Cost-of-living interpretation (10 minutes)

Inflation is an economy-wide average, but cost-of-living pressure can be felt very differently across households. Use your charts to anchor your answers.

**D1. Canadian cost-of-living channels.** Pick **two** Canadian-relevant channels and write one paragraph on each, connecting them to your inflation surge period:

- food and groceries,
- housing costs (rent, mortgage interest),
- transportation and gasoline,
- services inflation,
- exchange rate and imported goods.

**Channel 1 paragraph:**

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**Channel 2 paragraph:**

## Part E. Optional extension (5 minutes)

Compute an approximate **real policy rate**:

$$r_t \approx i_t - \pi_t,$$

where $i_t$ is the policy rate and $\pi_t$ is year-over-year inflation.

**E1.** Identify one period where the approximate real policy rate is negative. In Canada, who is more likely to benefit from negative real rates (borrowers or savers), and why?

## 2-minute group share-out (if time permits)

Each group shares: (i) the month of peak inflation, (ii) whether policy appears to lag or lead, and (iii) one Canadian cost-of-living channel that is most salient in your charts.